Unit 1

1. Introduction to AI

Tutorial 1

1. Introduction to AI
   1.1 Definition of AI and AI Effect
   1.2 Narrow, General and Super
   1.3 AI-Based and Conventional Systems.
   1.4 AI Technologies
   1.5 AI Development Frameworks
   1.6 Hardware for AI-Based Systems
   1.7 AI as a Service (AIaaS)
   1.8 Pre-Trained Models
   1.9 Standards, Regulations and AI

Introduction to AI

- Artificial Intelligence (AI) is the field of computer science which focuses on creating systems that can perform tasks which typically require human intelligence
- Few Core Concepts of AI are:
  - Machine Learning (ML):
    - Algorithms that allow systems to learn from data and improve over time without being explicitly programmed.
  - Deep Learning:
    - Deep learning is a subset of ML which uses neural networks to learn from complex patterns from huge data set.
  - Natural Language Processing (NLP):
    - NLP enables machines to understand and generate human language
  - Generative AI (GenAI):
    - Subset of ML/DL which can create new, original like content.
  - Computer Vision:
    - Computer vision allows machines to interpret and analyze visual information

Few Real-World Applications of AI

- Entertainment:
  - Movie and music recommendation
  - Generating creative content
- Customer Service:
  - Chatbots
  - Virtual assistants like Google Assistant, Siri and Alexa
- Finance:
  - Fraud detection
  - Automated trading
- Transportation:
  - Powering self-driving cars

THANK YOU!!!